

Entries

ATLAS Simulation Internal

$\sqrt{s} = 14$ TeV, HL-LHC, VBF $H \rightarrow \text{inv.}$

HGTD

$\sigma_g = 13.00, \sigma_d = 30.14$

$\alpha = 2.27, n = 1.17$

$\chi^2/\text{ndf} = 337.1/274$

$(S+M)/B = 90.13$

- Signal (>75%)
- Mixed (50–75%)
- Background (<50%)
- Gauss+DSCB Fit

Δt [ps]

10^4

10^3

10^2

10

1

-400

-300

-200

-100

0

100

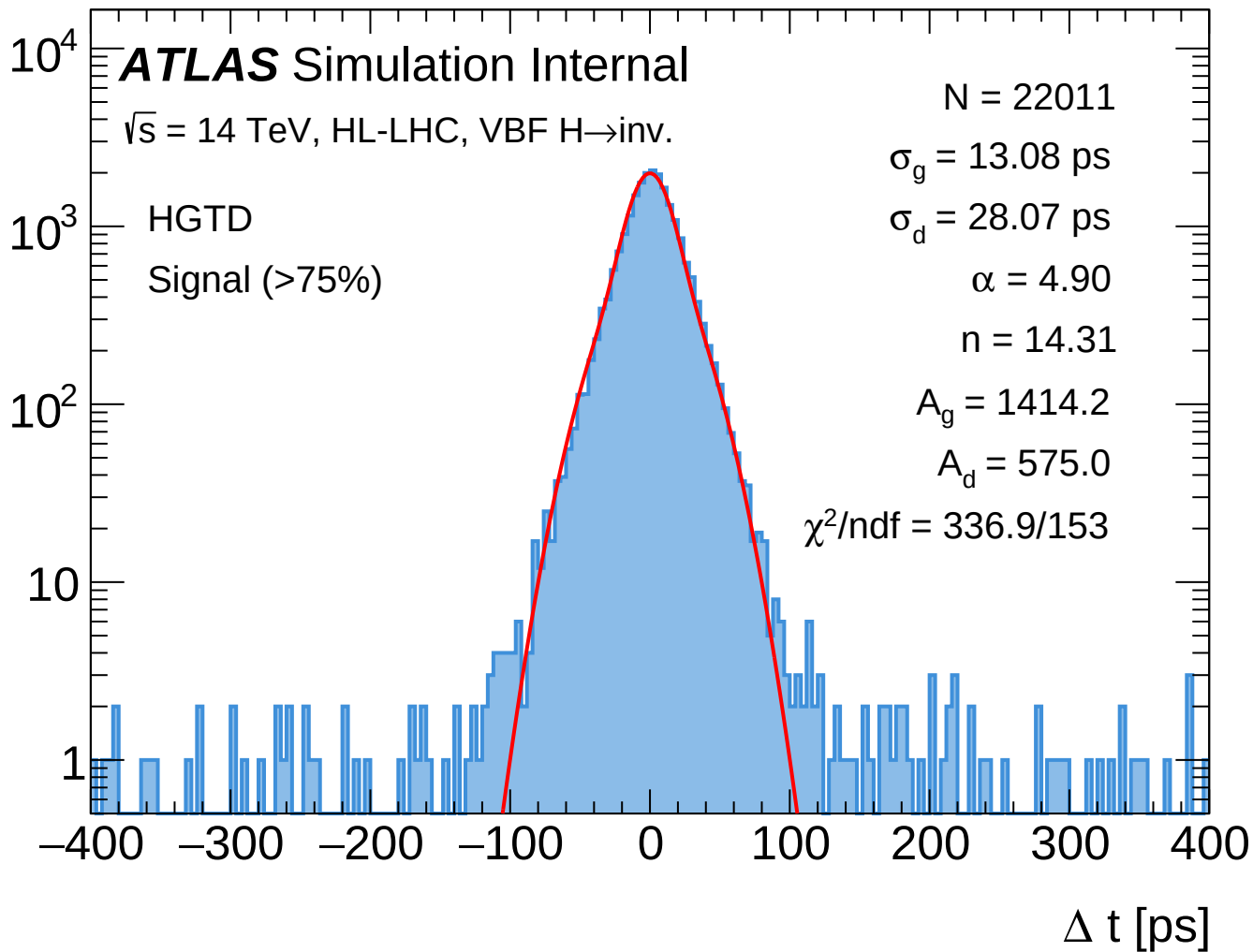
200

300

400

Δt [ps]

Entries



Entries

ATLAS Simulation Internal

$\sqrt{s} = 14$ TeV, HL-LHC, VBF $H \rightarrow \text{inv.}$

HGTD

Mixed (50–75%)

$N = 5440$

$\sigma_g = 15.77$ ps

$\sigma_d = 33.27$ ps

$\alpha = 2.32$

$n = 1.01$

$A_g = 237.4$

$A_d = 138.9$

$\chi^2/\text{ndf} = 107.7/141$

—400 —300 —200 —100 0 100 200 300 400

Δt [ps]

10^3

10^2

10

1

Entries

ATLAS Simulation Internal

$\sqrt{s} = 14$ TeV, HL-LHC, VBF $H \rightarrow \text{inv.}$

HGTD

Background (<50%)

$N = 2927$

$\sigma_g = 6.92$ ps

$\sigma_d = 42.60$ ps

$\alpha = 1.72$

$n = 1.01$

$A_g = 32.3$

$A_d = 83.7$

$\chi^2/\text{ndf} = 162.2/240$

-400 -300 -200 -100 0 100 200 300 400

Δt [ps]

10^2

10

1

Entries

ATLAS Simulation Internal

$\sqrt{s} = 14$ TeV, HL-LHC, VBF $H \rightarrow \text{inv.}$

TRKPTZ

$\sigma_g = 10.22, \sigma_d = 24.37$

$\alpha = 2.07, n = 1.06$

$\chi^2/\text{ndf} = 1067.5/300$

$(S+M)/B = 88.36$

- Signal (>75%)
- Mixed (50–75%)
- Background (<50%)
- Gauss+DSCB Fit

Δt [ps]

10^4

10^3

10^2

10

1

-400

-300

-200

-100

0

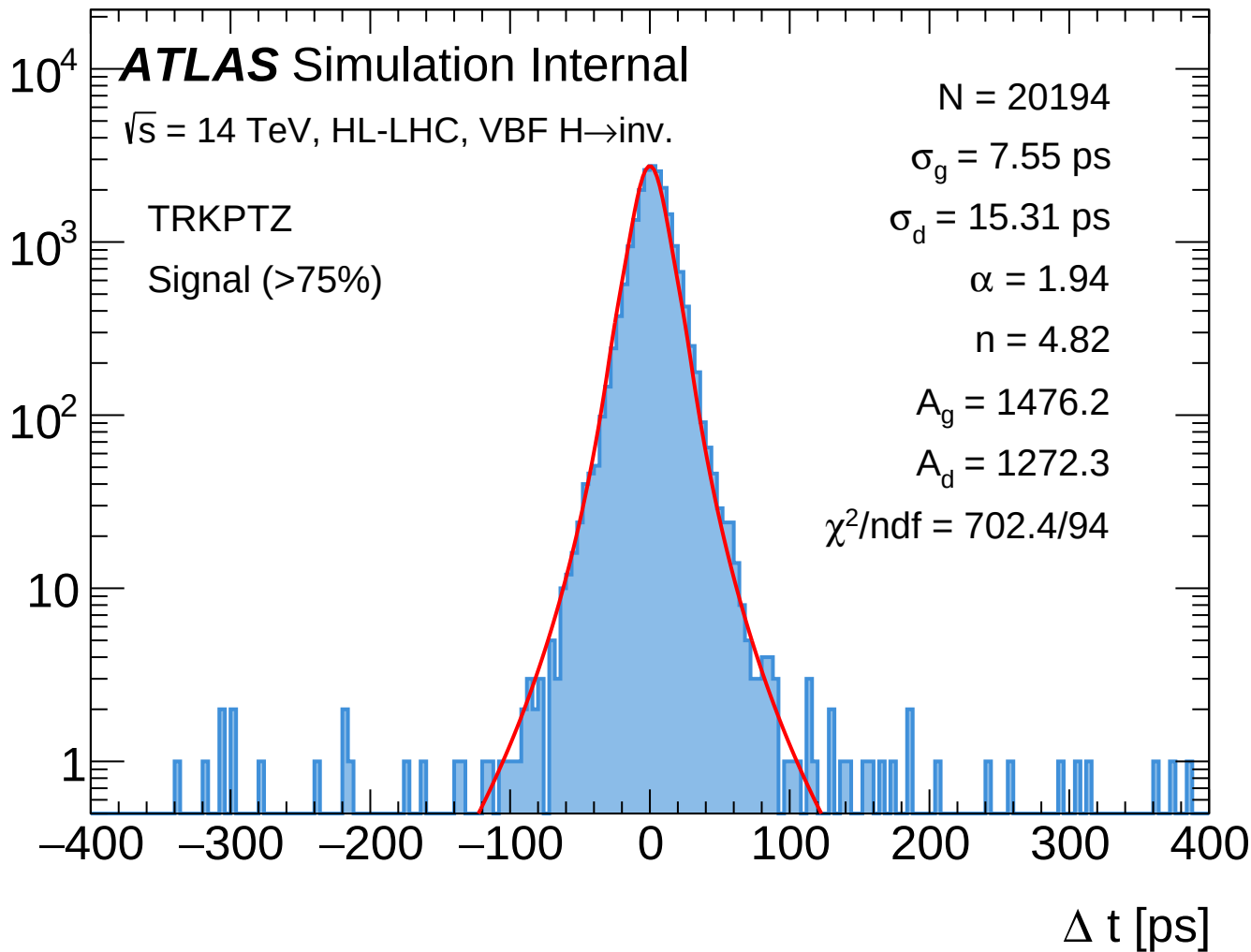
100

200

300

400

Entries



Entries

ATLAS Simulation Internal

$\sqrt{s} = 14$ TeV, HL-LHC, VBF $H \rightarrow \text{inv.}$

TRKPTZ

Mixed (50–75%)

$N = 7877$

$\sigma_g = 13.28$ ps

$\sigma_d = 30.01$ ps

$\alpha = 2.32$

$n = 1.01$

$A_g = 495.4$

$A_d = 184.3$

$\chi^2/\text{ndf} = 182.8/128$

10^3

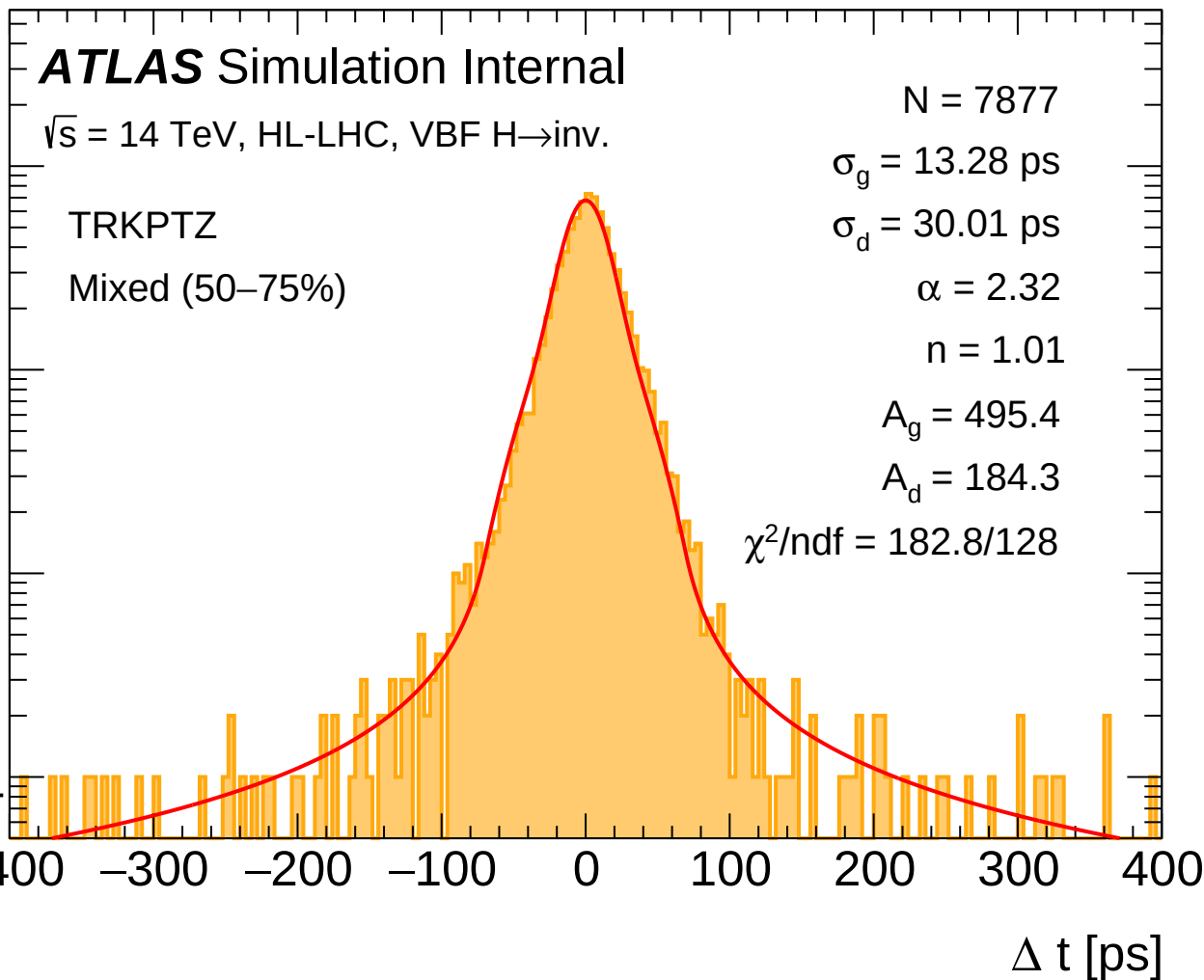
10^2

10

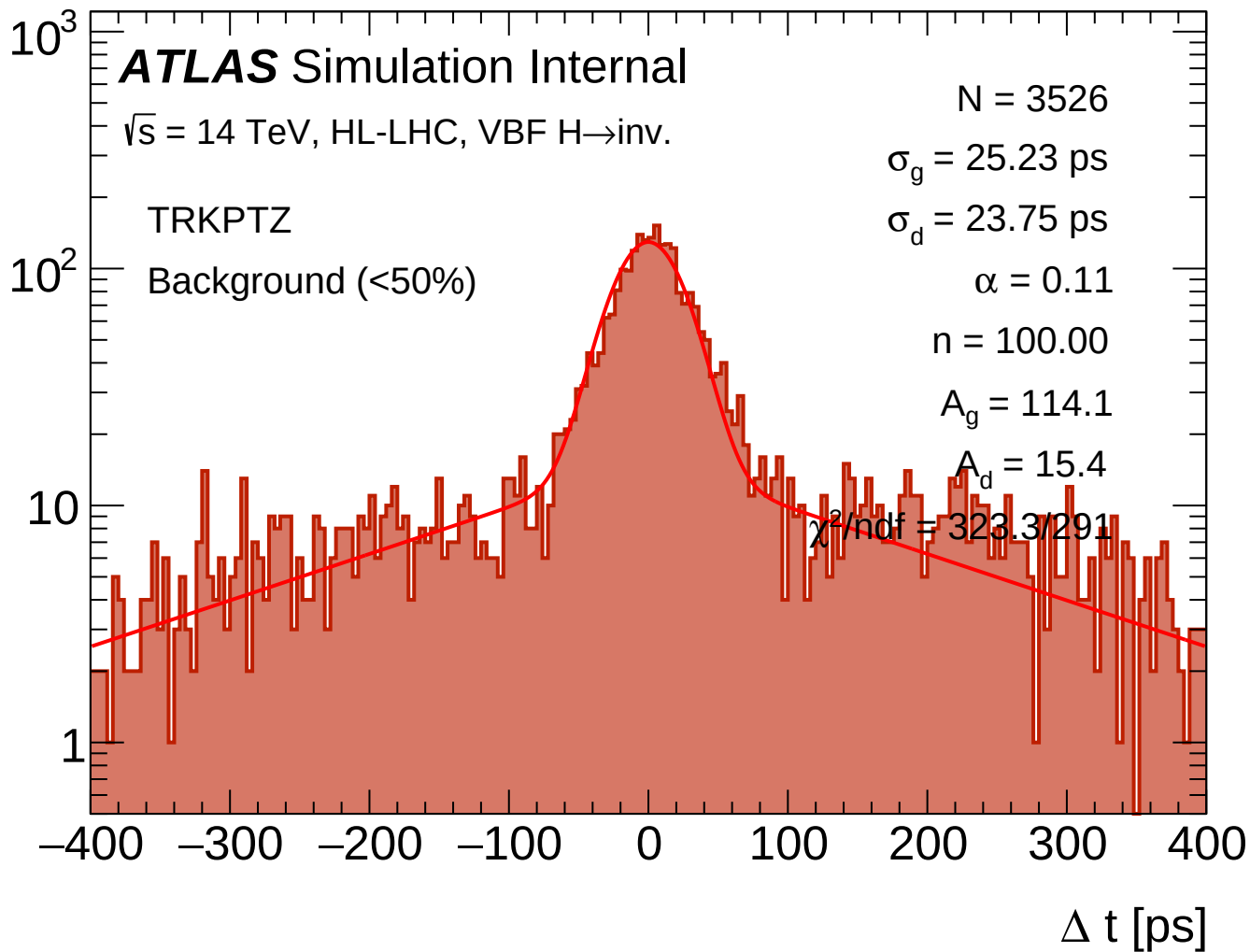
1

–400 –300 –200 –100 0 100 200 300 400

Δt [ps]



Entries



Entries

ATLAS Simulation Internal

$\sqrt{s} = 14$ TeV, HL-LHC, VBF $H \rightarrow \text{inv.}$

TRKPTZ Pure Clust.

$\sigma_g = 7.55, \sigma_d = 15.31$

$\alpha = 1.94, n = 4.82$

$\chi^2/\text{ndf} = 702.4/94$

$(S+M)/B = 100.00$

- Signal (>75%)
- Mixed (50–75%)
- Background (<50%)
- Gauss+DSCB Fit

Δt [ps]

10^4

10^3

10^2

10

1

-400

-300

-200

-100

0

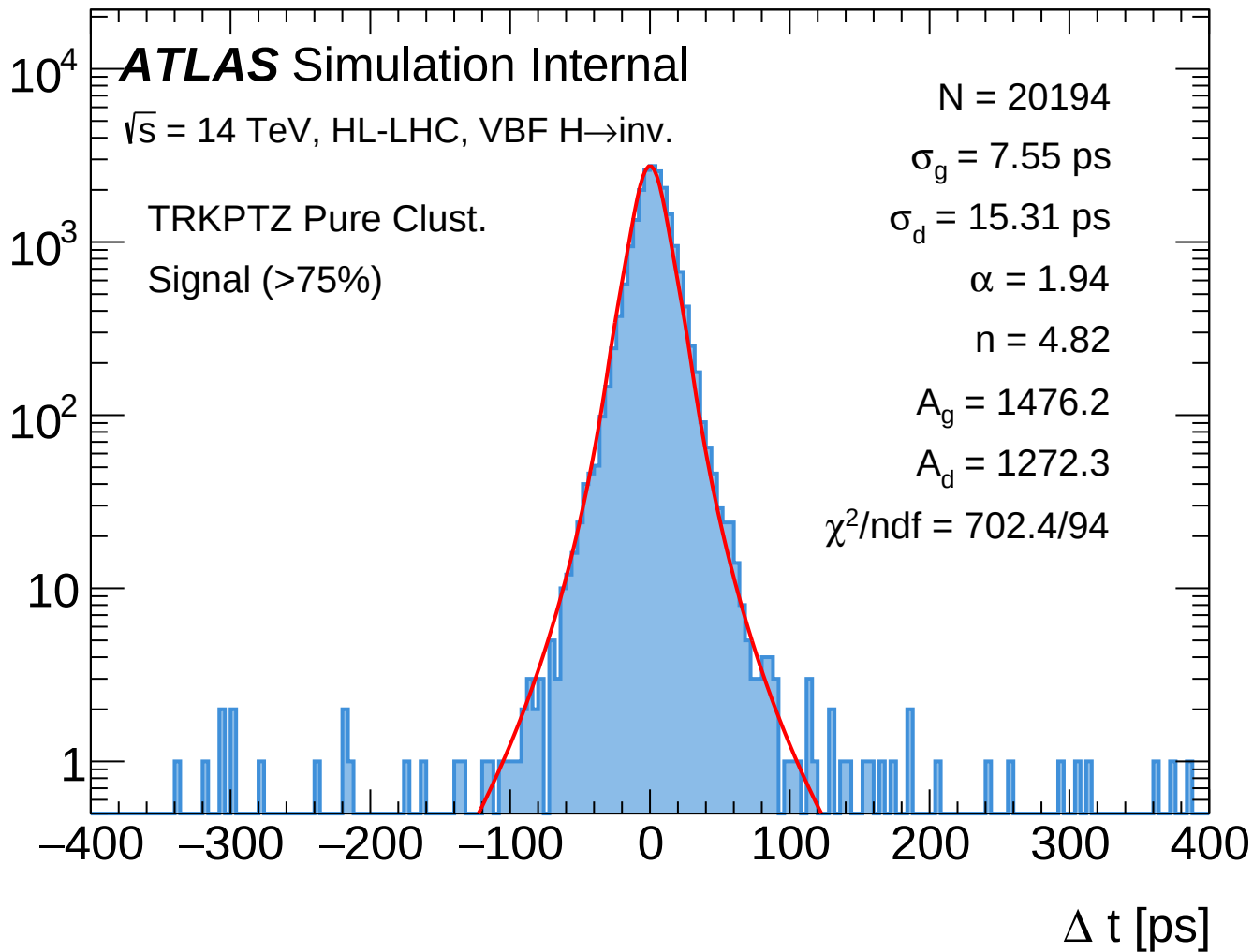
100

200

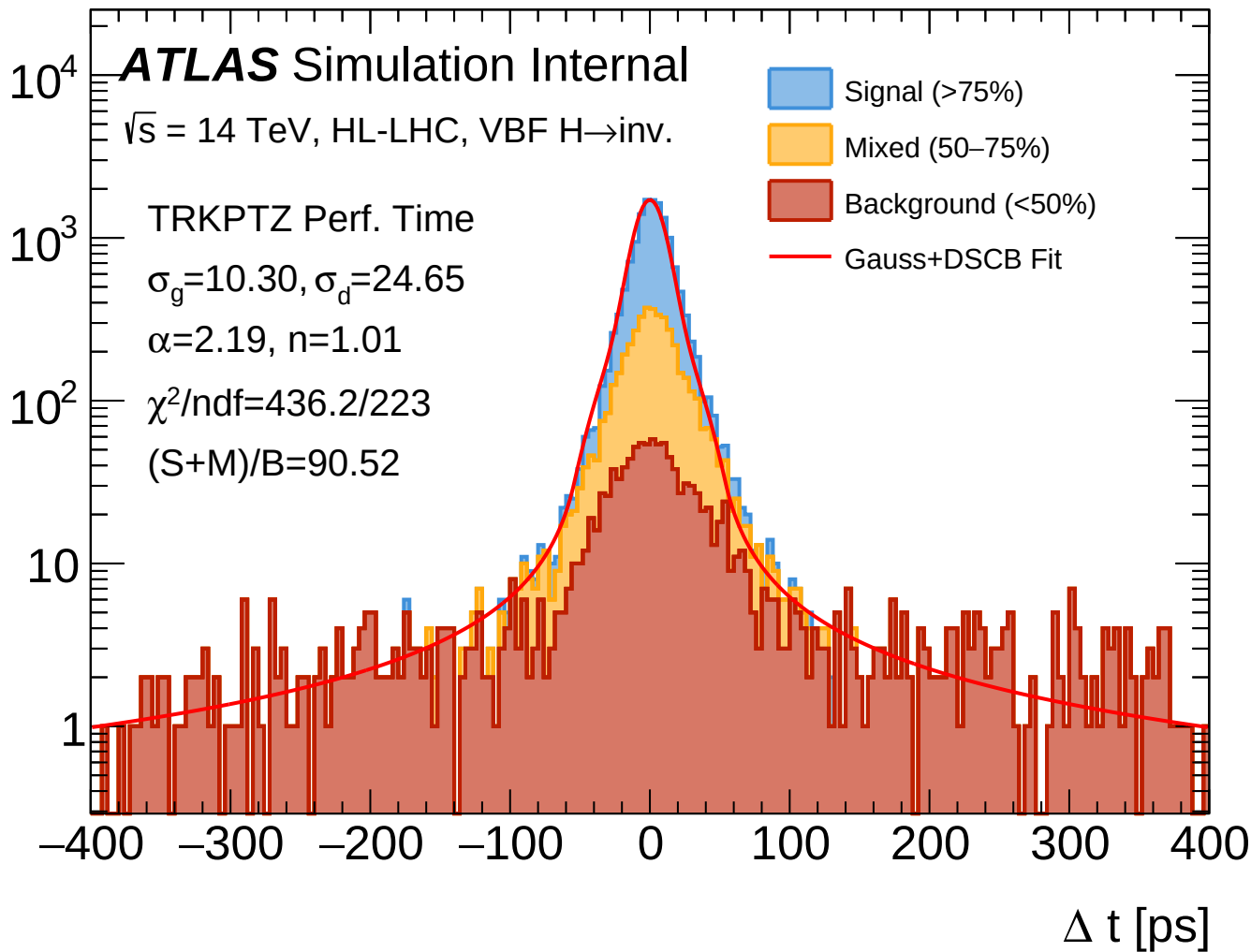
300

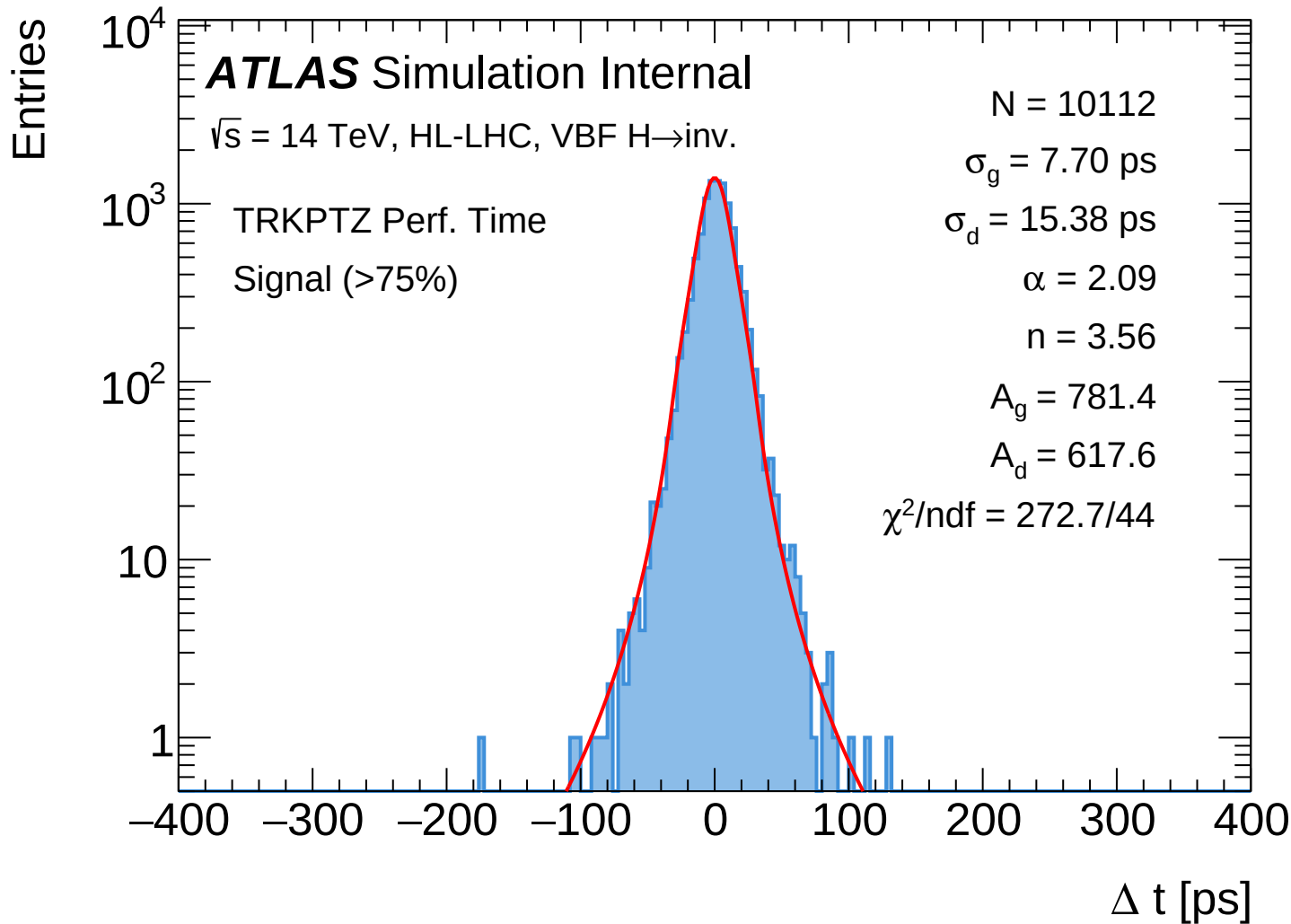
400

Entries



Entries





Entries

ATLAS Simulation Internal

$\sqrt{s} = 14$ TeV, HL-LHC, VBF $H \rightarrow \text{inv.}$

TRKPTZ Perf. Time

Mixed (50–75%)

$N = 3547$

$\sigma_g = 12.98$ ps

$\sigma_d = 28.32$ ps

$\alpha = 2.03$

$n = 1.59$

$A_g = 210.8$

$A_d = 96.2$

$\chi^2/\text{ndf} = 76.9/56$

10^3

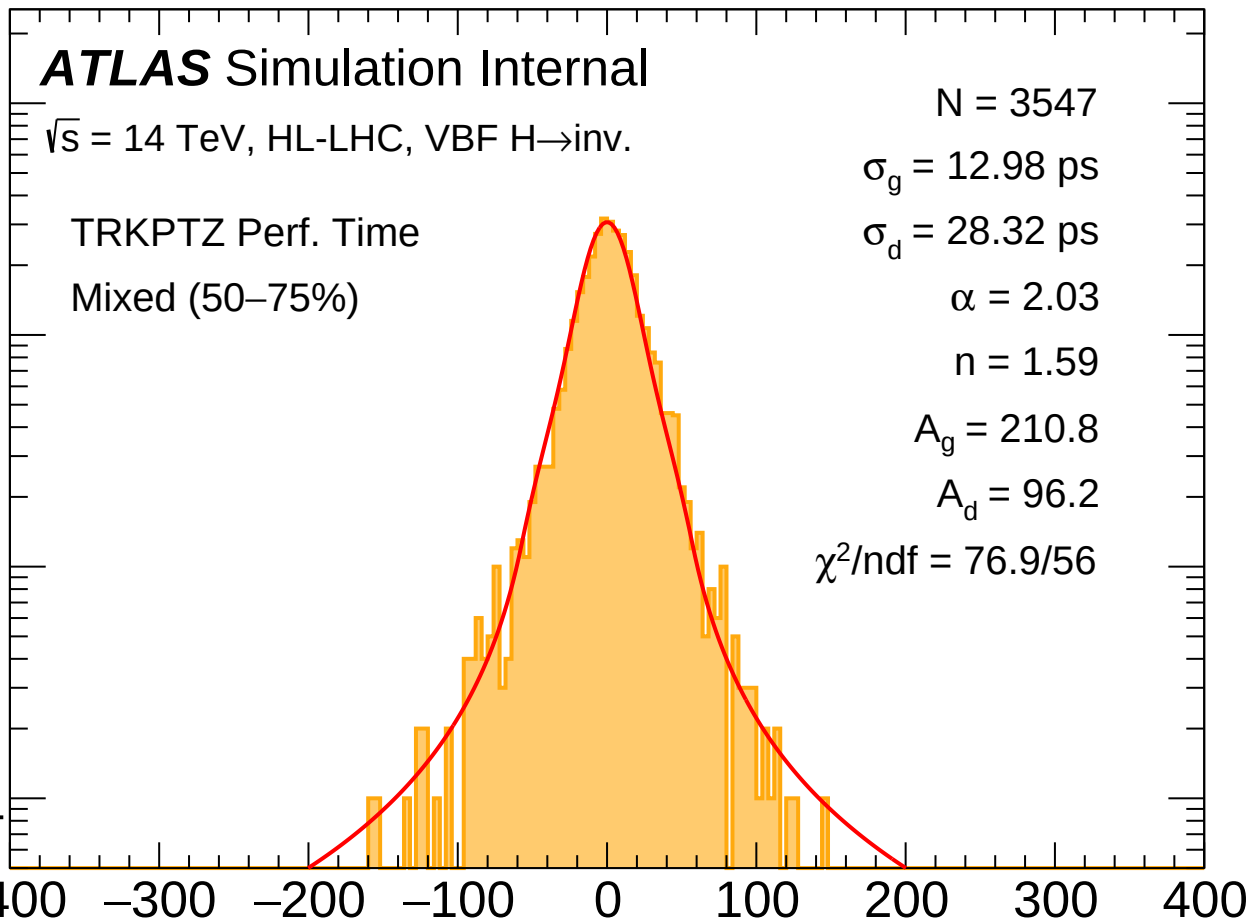
10^2

10

1

–400 –300 –200 –100 0 100 200 300 400

Δt [ps]



Entries

ATLAS Simulation Internal

$\sqrt{s} = 14$ TeV, HL-LHC, VBF $H \rightarrow \text{inv.}$

TRKPTZ Perf. Time

Background (<50%)

$N = 1373$

$\sigma_g = 26.65$ ps

$\sigma_d = 20.65$ ps

$\alpha = 0.11$

$n = 1.21$

$A_g = 47.2$

$A_d = 4.6$

$\chi^2/\text{ndf} = 138.9/223$

Δt [ps]

10^2

10

1

-400

-300

-200

-100

0

100

200

300

400

Entries

ATLAS Simulation Internal

$\sqrt{s} = 14$ TeV, HL-LHC, VBF $H \rightarrow \text{inv.}$

TEST_ML

$\sigma_g = 10.14, \sigma_d = 23.97$

$\alpha = 2.06, n = 1.07$

$\chi^2/\text{ndf} = 1001.3/302$

$(S+M)/B = 88.85$

- Signal (>75%)
- Mixed (50–75%)
- Background (<50%)
- Gauss+DSCB Fit

—400 —300 —200 —100 0 100 200 300 400

Δt [ps]

10^4

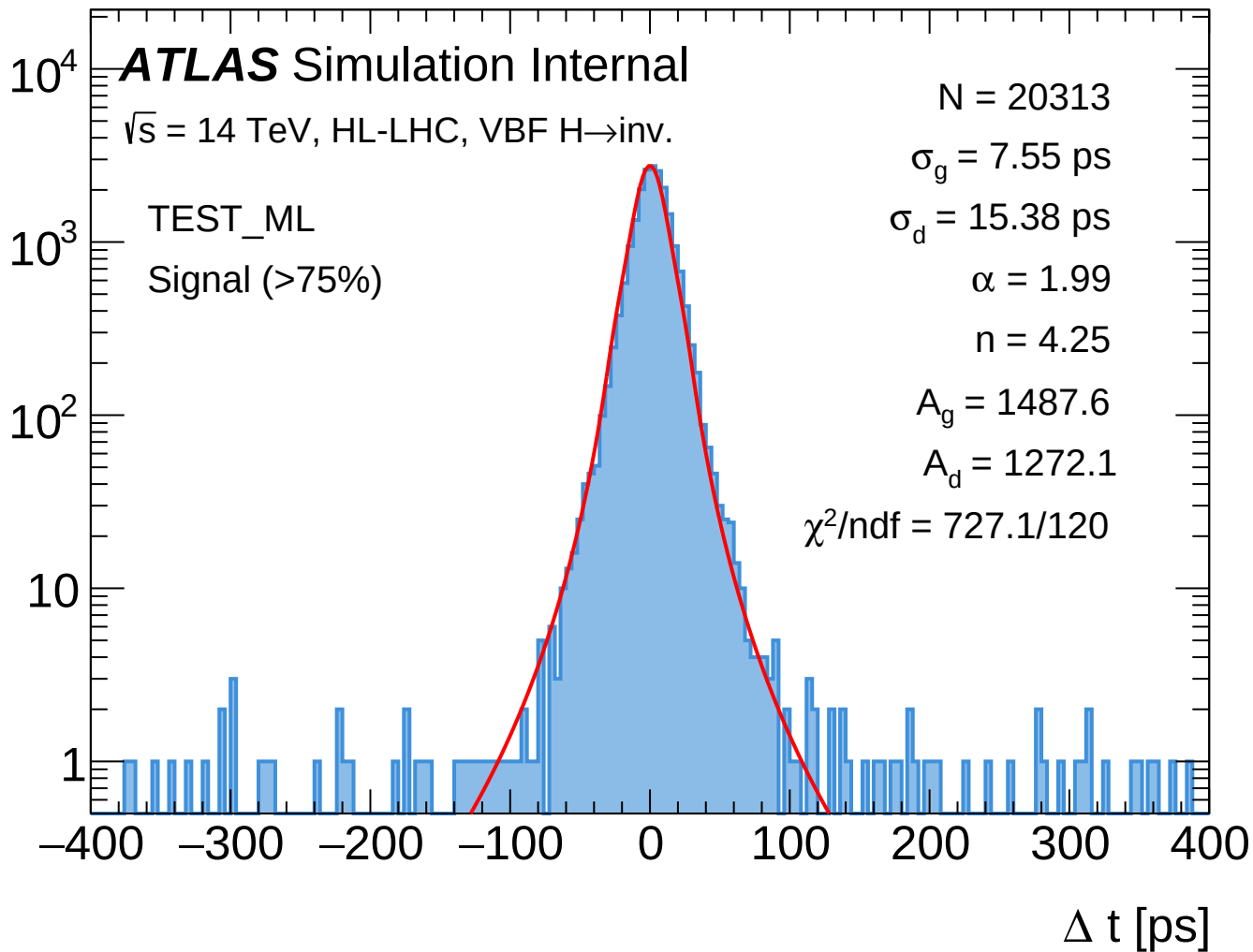
10^3

10^2

10

1

Entries



Entries

ATLAS Simulation Internal

$\sqrt{s} = 14$ TeV, HL-LHC, VBF $H \rightarrow \text{inv.}$

TEST_ML

Mixed (50–75%)

$N = 7909$

$\sigma_g = 13.58$ ps

$\sigma_d = 30.78$ ps

$\alpha = 2.30$

$n = 1.01$

$A_g = 502.5$

$A_d = 172.2$

$\chi^2/\text{ndf} = 198.6/146$

10^3

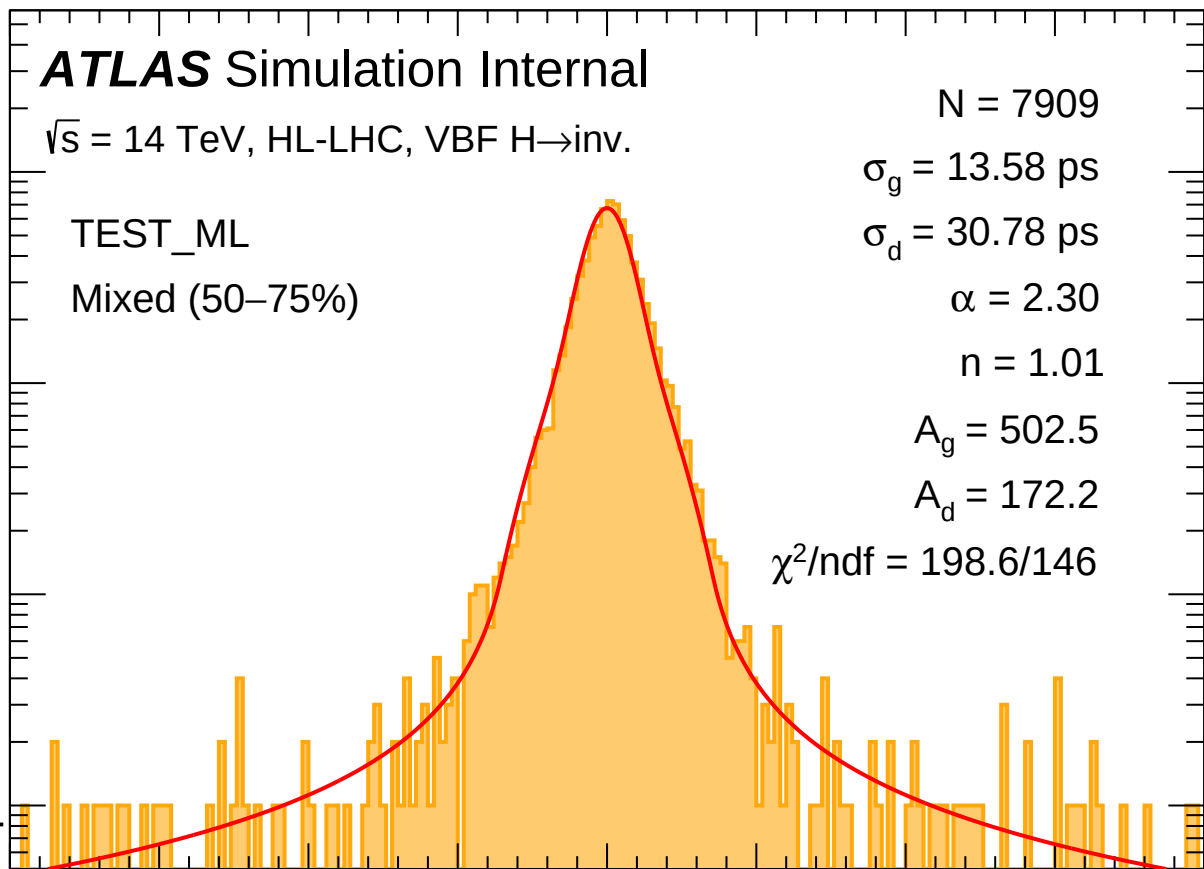
10^2

10

1

–400 –300 –200 –100 0 100 200 300 400

Δt [ps]



Entries

